

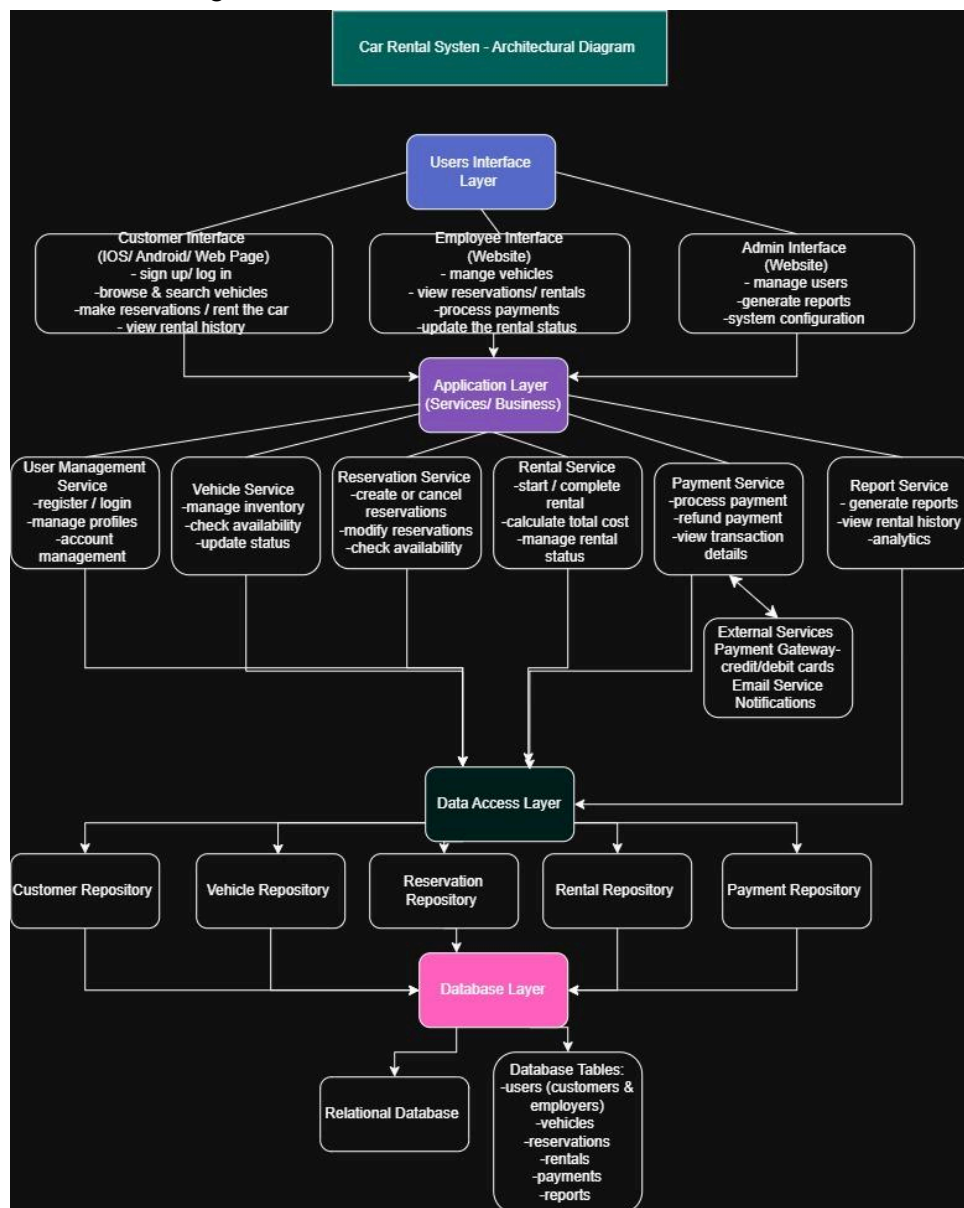
## BeAvis Car Rental System

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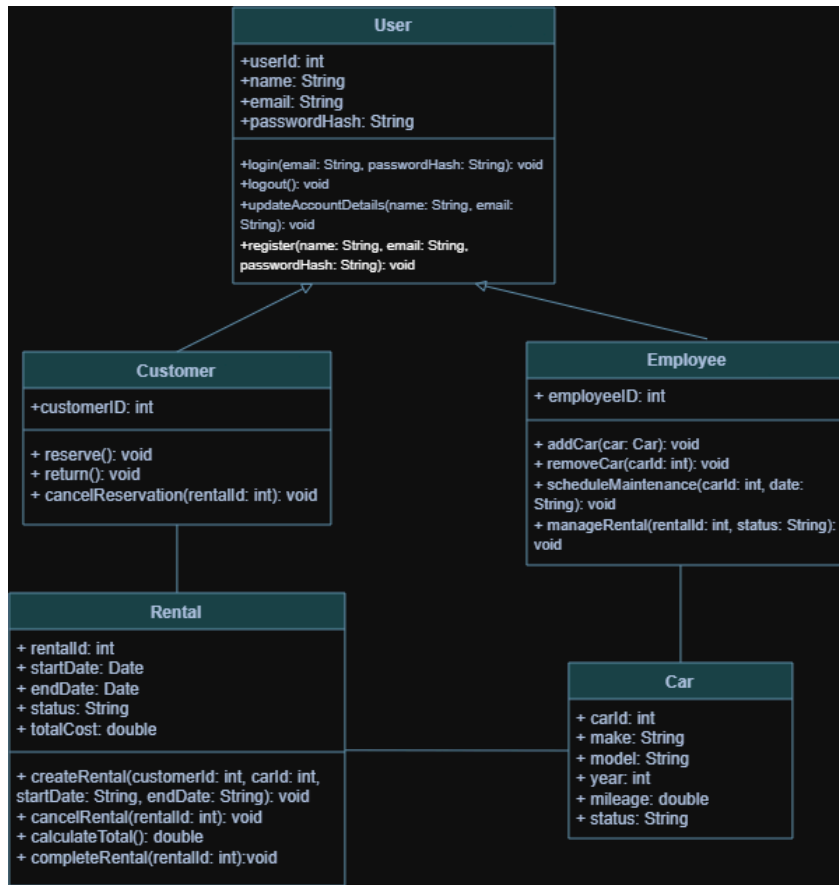
This is a car rental system designed to manage a car rental company. This system will allow customers to rent vehicles both online and via a mobile app provided they have signed up for an account with the company. This system will also facilitate the distribution and signing of rental agreement contracts under secure conditions.

Employees will also be able to access the system in order to look up customers. Upon looking up a customer on the system, they will be able to see the rental status and contracts, check what cars are currently available, and update the status of cars.

Architectural Diagram:



UML Diagram:



## User class

### Description

The User class represents a general person who uses the BeAvis system. It is the parent class for the Employee and Customer classes. This class stores common account information and allows users to create accounts, log in, log out, and update their personal details. Passwords should not be stored as plain text. Instead, the system should store a secure password hash.

### Attributes

- `userId: int`
- `name: String`
- `email: String`
- `passwordHash: String`

### Operations

- `register(name: String, email: String, password: String): boolean`
- `login(email: String, password: String): boolean`
- `logout(): void`
- `updateAccountDetails(name: String, email: String): void`

## Employee Class

### Description

The Employee class represents a BeAvis employee; it inherits the common account information and operations from the user class. Employees have additional permissions because they manage vehicles,

customers, rentals, and maintenance information. Employees can update the availability of cars and review customers rental information

#### **Attributes**

- employeeId: int

#### **Operations**

addCar(car: Car): void

removeCar(carId: int): boolean

scheduleMaintenance(carId: int, description: String): void

manageRental(rentalId: int, action: String): boolean

#### **Rental Class**

##### **Description**

The rental class represents a vehicle reservation made by a customer. It stores information about the rental dates, rental status and total price. A rental is connected to one customer and one car. The class allows the system to create, cancel, calculate and complete rental transactions.

##### **Attributes**

- rentalId: int
- startDate: Date
- endDate: Date
- status: String
- totalCost: double

##### **Operations**

- + createRental(customerId: int, carId: int, startDate: Date, endDate: Date): Rental
- + cancelRental(): boolean
- + calculateTotal(dailyRate: double): double
- + completeRental(): void

#### **Car Class**

##### **Description**

The car class represents a vehicle in the BeAvis inventory, it stores information about the car including model, year, mileage and current status. The car status shows whether the vehicle is available for reserved, rented, or under maintenance

##### **Attributes**

- carId: int
- make: String
- model: String
- year: int
- mileage: double
- status: String

##### **Operations**

updateStatus(newStatus: String): void

getDetails(): String

isAvailable(startDate: Date, endDate: Date): boolean

Title page: Christopher

Software title: Christopher

Team members: Christopher

System Description: Christopher

Brief overview of system: Christopher

Software Architecture Overview

Architectural diagram of all major components: Michelle

UML Class Diagram: Kevin

\* descriptions should be detailed and specify datatypes, function interfaces, parameters, etc..

Development plan and timeline : Light

Partitioning of tasks : Light

Team member responsibilities: Light